Exploring the Stroop Effect of Object-Color Association

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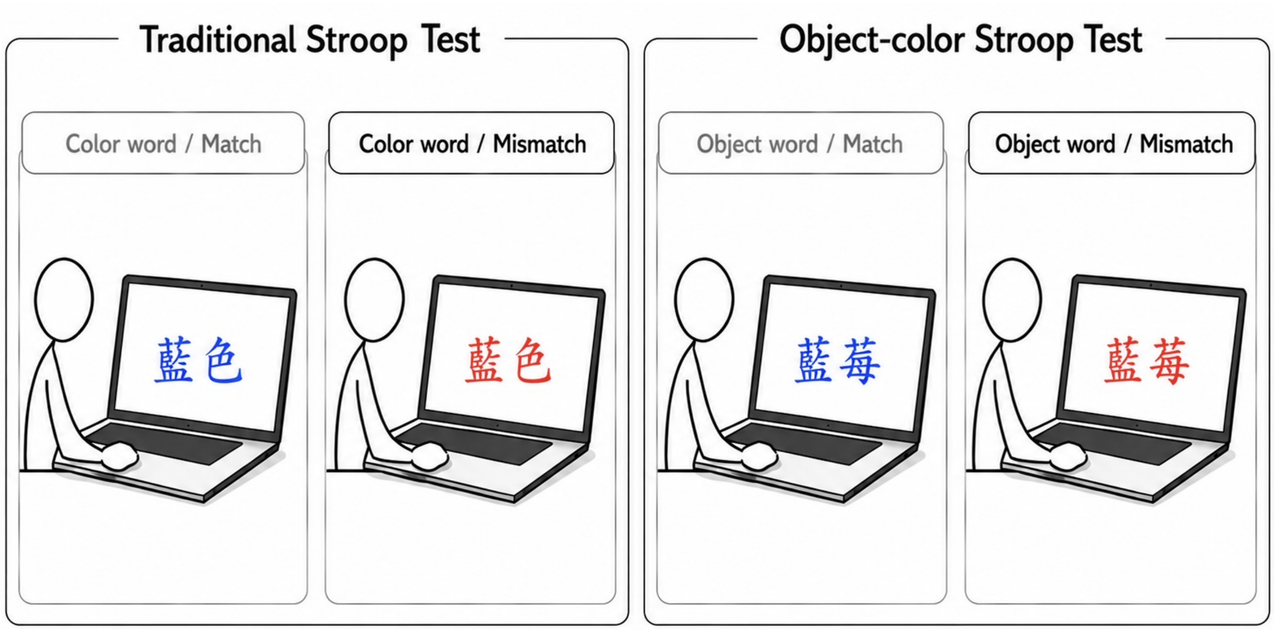

Figure 1: Overview of the four conditions used in the experiment.

Abstract

This study examines whether object-color associations can produce a Stroop-like interference effect in a visual color judgment task. While the classic Stroop effect shows that color-word meaning interferes with font color identification, this study extends the paradigm to object words with strong typical color associations, such as fruit words. Participants were asked to ignore word meaning and identify the actual font color as quickly and accurately as possible under match and mismatch conditions. The results showed better performance in match conditions than in mismatch conditions, indicating that incongruent semantic information interfered with color judgment. Notably, compared with color-mismatch trials, fruit-mismatch trials led to lower accuracy but shorter reaction times, suggesting that typical object-color associations may prompt participants to respond quickly but may also lead to incorrect judgments. Overall, the findings suggest that object color knowledge can be automatically activated and affect visual decision-making.

Keywords

Stroop effect, object-color association, color perception

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1 Introduction

Color is one of the important features in visual recognition. Previous studies have shown that color information can contribute to object recognition, especially when an object has a stable and familiar association with a particular color [2, 15]. In everyday life, people often associate many objects with specific colors. For example, apples are commonly associated with red, the sky with blue, and the sun with yellow. These object-color associations are usually formed through long-term daily experience. Therefore, when people see or read about an object, they may automatically activate the typical color associated with that object. Prior research on memory color and object-color knowledge also suggests that stored color knowledge can influence color perception and may be activated rapidly during visual processing [5, 12].

A classic phenomenon in cognitive psychology, the Stroop effect, demonstrates the interference between semantic information and color information [14]. In a traditional Stroop task, when the meaning of a word is inconsistent with its font color, participants usually take longer to correctly identify the font color. In other words, the semantic meaning of the word becomes a source of interference during color judgment. Therefore, the Stroop effect has been widely used as a paradigm for studying selective attention and cognitive interference [8, 13]. Although the traditional Stroop task mainly focuses on the conflict between color words and font colors, studies on color diagnosticity and object-color representation suggest that ordinary objects may be linked to stable color knowledge and may activate their typical colors during visual processing [10, 12, 16, 18]. This effect may be especially strong for common objects in everyday life, as they tend to form stable and familiar associations with specific colors. Fruits are representative examples of such stimuli because they are colorful and frequently encountered in daily life, and previous work has shown that fruit-related object memories are strongly linked to color information [7]. For instance, apples are commonly associated with red, watermelons with green, mangoes with yellow, blueberries with blue, and grapes with purple.

Therefore, this experiment investigates whether typical colors associated with objects influence users' reaction time and accuracy. We adopted a visual-only design using five colors and five fruit words as stimuli. The trials were divided into match and mismatch conditions: in the match condition, the displayed font color was consistent with the meaning of the color word or the typical color of the fruit word, whereas in the mismatch condition, it was inconsistent with them. By comparing reaction time and accuracy across these conditions, this study examines whether object-related color knowledge can automatically influence users' responses.

We hypothesize that participants will respond faster and more accurately in match trials than in mismatch trials. If mismatch trials result in longer reaction times or lower accuracy, this would suggest that the typical color associated with a fruit word may be automatically activated and interfere with font color judgment. Therefore, this experiment extends the traditional Stroop paradigm from color-word conflict to the relationship between everyday objects and their typical color associations.

2 Related Work

2.1 Stroop Effect and Its Extensions

The Stroop effect is a classic phenomenon in cognitive psychology and has been widely used to study selective attention and cognitive interference. In the original Stroop task, participants named the ink color of a printed word while ignoring its semantic meaning. When word meaning and ink color were incongruent, participants typically responded more slowly and made more errors [14]. Later reviews showed that the Stroop effect provides an important paradigm for understanding how automatic word processing interferes with task-relevant color judgment [8, 13]. This phenomenon has also been described as interdimensional interference, in which different stimulus dimensions, such as semantic meaning and visual color, compete during attention and response selection [9]. Recent work further suggests that Stroop effects may arise at multiple processing stages, including semantic processing, attentional selection, and response selection [11]. These studies show that Stroop interference can occur when semantic information that should be ignored competes with color judgment.

2.2 Color Diagnosticity in Object Recognition

Beyond the traditional Stroop task, research has examined color in object recognition. Early work suggested that shape or edge-based information may play a primary role, while surface features such as color may play a secondary role [1]. Later studies, however, showed that color can contribute to object recognition, especially for natural or familiar objects whose colors are strongly associated with their identities [6, 15]. This idea is related to color diagnosticity, which refers to how strongly an object is associated with a specific color. Color information has been shown to have a stronger effect on high color-diagnostic objects, such as fruits and vegetables, than on objects with less fixed color associations [16]. Other studies also suggest that stored color knowledge is part of object representation and can influence object processing [2, 3, 18]. Research on fruit-related object memory further shows that color is strongly linked to fruit object representation [7]. These findings make fruit words suitable stimuli for examining object-color associations.

2.3 Object Color Knowledge

Object color knowledge may not only support object recognition but also influence color judgment. A Stroop-like color-naming study showed that participants responded faster when objects appeared in typical colors than in atypical colors [10], indicating that object-related color knowledge can produce facilitation or interference during color judgment. Research on memory color further showed that prior knowledge of an object's typical color can modulate perceived color appearance [5]. Neurocognitive studies also suggest that prior object knowledge can shape color representation in the visual cortex and influence emerging object representations in the brain [17, 19]. Recent work has also reported automatic and early color-specific neural responses to object color knowledge, supporting the idea that typical color associations may be rapidly activated during object processing [12]. Taken together, these studies suggest that semantic-color conflict is not limited to direct color-word interference; familiar object-color associations may also interfere with visual color judgment. Therefore, the present study examines whether fruit words can produce Stroop-like interference when their typical colors are inconsistent with the displayed font color.

3 User Study

3.1 Study Design

This study aims to investigate whether the association between objects and colors influences participants' performance in a color judgment task. In each trial, participants were required to respond based on the actual font color of the displayed word, rather than the semantic meaning of the word itself.

The experiment included two manipulated variables. The first independent variable was the type of visual stimulus, which was divided into color words and object words. In color-word trials, the screen presented color-related words, such as red, green, orange, blue, and purple. In object-word trials, the screen presented object names. Since previous research has shown that fruits are strongly associated with colors, this study used fruits as the object-word stimuli, including **apple, watermelon, mango, blueberry, and grape**. The second independent variable was congruency, which was divided into match and mismatch conditions. In the match condition, the semantic meaning of the word was consistent with the actual font color. For example, when the displayed word was "red," the font color was also red. Similarly, when the displayed word was "apple," the font color was red. In the mismatch condition, the semantic meaning of the word was inconsistent with the actual font color. For example, when the displayed word was "red," the font color could be blue, as shown in Figure 1.

The experiment consisted of four experimental conditions: color-match, color-mismatch, fruit-match, and fruit-mismatch. Since the experiment included five target colors and four experimental conditions, each color-condition combination appeared three times. Therefore, each participant completed a total of 60 formal trials. To reduce the influence of order effects, all trials were presented in a randomized order. In addition, the trial sequence was designed to avoid consecutive trials with the same correct answer or the same trial category whenever possible. This reduced the likelihood that participants would develop prediction-based or habitual responses due to highly similar consecutive stimuli.

The dependent variables were reaction time and accuracy. Reaction time was defined as the time interval between the onset of the visual stimulus and the participant's key press. Accuracy was defined as whether the participant pressed the correct key corresponding to the actual font color. We hypothesized that mismatch trials would result in longer reaction times and lower accuracy than match trials because the semantic meaning of the word may interfere with participants' judgment of the font color. We further hypothesized that fruit-mismatch trials may also produce interference, suggesting that typical object-color associations can influence color judgment even when the stimulus is not a direct color word.

3.2 Procedure

The experiment was conducted in a quiet environment. Before the trials, participants read the task instructions and were told to respond based on the font color of each displayed word rather than its semantic meaning. They then completed a fruit-color matching task to confirm that their fruit-color associations were consistent with the stimulus design and to familiarize themselves with the color categories and response method. During the experiment, a fixation cross was shown between trials as a brief rest and attention cue. In each trial, a word stimulus appeared in one of the five target colors, and participants judged its font color as quickly and accurately as possible by pressing the corresponding key. After each response, the system recorded the selected key and reaction time before proceeding to the next trial. After all trials were completed, the response data were saved and used to calculate the average reaction time and accuracy for each experimental condition.

3.3 Participants

A total of 23 participants were recruited, mainly from laboratory members and university classmates. All had normal or corrected-to-normal vision, could distinguish the five colors used in the study, and were native Chinese speakers or long-term primary Chinese users. This ensured that they could process the Chinese word stimuli naturally. The participants included 7 females and 16 males, with ages ranging from 20 to 25 years old.

3.4 Apparatus

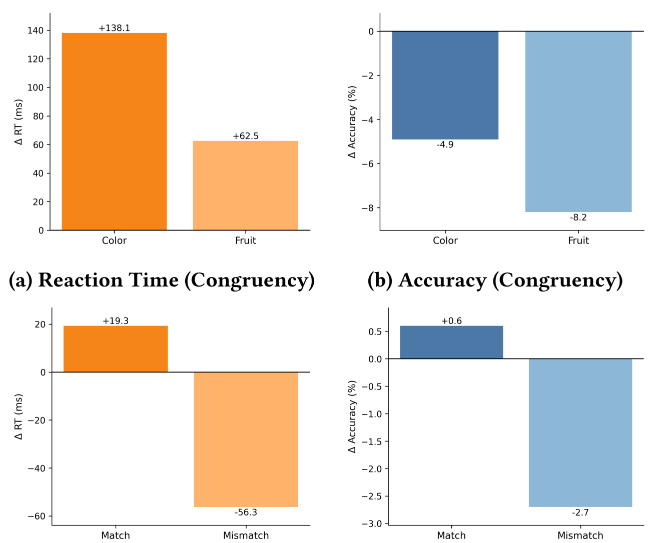
The experiment was conducted on a laptop computer, and participants responded using a keyboard. To support efficient responses and accurate reaction time recording, each color was assigned to a fixed key: R for red, G for green, O for orange, B for blue, and P for purple. Participants pressed the key corresponding to the actual font color of the word rather than its semantic meaning. For each trial, the system recorded the stimulus type, congruency condition, semantic color, actual font color, participant response, reaction time, and correctness. Reaction time was recorded in milliseconds.

4 Results

Table 1 presents the mean accuracy and reaction time across the four experimental conditions. To further examine how different factors influenced task performance, Figure 2 visualizes the delta analysis for both the congruency effect and the word type effect. The congruency effect was computed as Mismatch minus Match within each word type, whereas the word type effect was computed as Fruit minus Color within each congruency condition.

Table 1: Mean Accuracy and Reaction Time by Condition.

Type	Congruency	Accuracy (%)	Reaction Time (ms)
Color	Match	97.9	1038.5
Color	Mismatch	93.0	1176.6
Fruit	Match	98.5	1057.8
Fruit	Mismatch	90.3	1120.3



(a) Reaction Time (Congruency) **(b) Accuracy (Congruency)**

(c) Reaction Time (Type) **(d) Accuracy (Type)**

Figure 2: Delta effects on reaction time and accuracy.

First, regarding the congruency effect, participants performed better in Match conditions than in Mismatch conditions. This suggests that semantic incongruency interfered with task performance. As shown in Figure 2a and 2b, both Color and Fruit trials showed lower accuracy and longer reaction time under the Mismatch condition. However, the Fruit condition showed a larger decrease in accuracy, while the Color condition showed a larger increase in reaction time. One possible interpretation is that color words induced stronger semantic conflict, causing participants to spend more time suppressing word meaning before responding. In contrast, fruit words may have activated typical object-color associations more rapidly, leading to faster but less accurate responses when the associated color conflicted with the actual font color.

Next, regarding the word type effect, the influence of word type differed between the Match and Mismatch conditions. As shown in Figures 2c and 2d, the differences between Color and Fruit trials in the Match condition were relatively small for both accuracy and reaction time. This suggests that when the semantic meaning or object-color association was consistent with the actual font color, the two word types did not produce substantially different effects on task performance. Although Fruit trials showed slightly longer reaction times and higher accuracy, the overall difference was limited, making it difficult to conclude that fruit words created a clear additional processing burden in the Match condition. In contrast, the difference between word types became more noticeable in the Mismatch condition. Fruit-Mismatch trials showed faster reaction times but lower accuracy than Color-Mismatch trials, suggesting a possible speed-accuracy trade-off. This pattern suggests that participants were quickly influenced by the typical color association of fruit words, which helped them respond faster but also increased the likelihood of incorrect responses when that associated color was inconsistent with the actual font color.

5 Limitations

This study has several limitations. First, the participant group was small and mainly consisted of young adult Chinese users with similar educational backgrounds, so the findings may not generalize to broader populations with different ages, cultures, or language experiences. Since Chinese word stimuli were used, future studies could examine whether similar effects occur in other languages or writing systems. Second, the stimuli were limited to five colors and five fruits. Although fruits are suitable high color-diagnostic objects, other objects may have weaker or more flexible color associations and produce different interference effects. Future work could include more object categories to examine how association strength influences reaction time and accuracy. Third, using a laptop keyboard may have introduced variation due to keyboard familiarity, typing speed, or key-search behavior. More controlled response devices or longer practice sessions could reduce this influence. Finally, this study only analyzed reaction time and accuracy, which cannot directly reveal the underlying cognitive process. Future research could combine behavioral data with eye-tracking or neurocognitive measures to better understand how object-color associations are activated and influence visual decision-making.

6 Potential Implications for User Interfaces

6.1 Implications for Learning Interfaces

These findings may inform educational or interactive interfaces for users with established object-color associations. For example, displaying the word “blueberry” in red during object or color identification tasks may interfere with users’ judgments because it conflicts with their established object-color associations. Consistent object-color pairings may reduce semantic conflict during visual tasks.

6.2 UI and Training Applications

Across user interfaces, designers should consider users’ common impressions of objects and their associated colors. Inconsistent object-color designs may increase cognitive load or lead users to make more errors during quick visual judgments. On the other hand, this inconsistency can be intentionally applied in games or interactive training systems. For example, in attention-training games or cognitive-control tasks, the system can present familiar objects in atypical colors, such as a red blueberry or a blue apple, and require users to suppress their automatic object-color associations while focusing on the displayed color. In this case, the mismatch between an object and its typical color is not only a design issue to be avoided, but can also serve as an interactive mechanism for training attention, response control, and cognitive inhibition.

7 Future Work

Future work could extend this experiment in several directions. First, a larger and more diverse participant group could be recruited to examine whether the observed effects remain stable across ages, cultural backgrounds, language experiences, and levels of object familiarity, since color may play different roles depending on familiarity and expertise [4]. Second, more object categories could be included to compare different levels of color diagnosticity and clarify whether stronger object-color associations are linked to stronger interference effects [16, 18]. Third, future studies could employ more controlled response devices, longer practice sessions, and eye-tracking techniques to reduce response variability and better capture participants’ visual attention during color judgment tasks. Finally, future studies could incorporate neurocognitive measures together with signal detection theory (SDT) to further investigate the cognitive mechanisms underlying the observed interference effects [12, 17, 19]. In particular, signal detection measures such as perceptual sensitivity and decision criterion may help distinguish whether the observed performance differences arise from semantic interference or shifts in decision strategy. These extensions could provide a more complete understanding of how implicit object-color knowledge influences perception and response behavior.

8 Conclusion

This study explored whether object-color associations can produce a Stroop-like interference effect in a visual color judgment task. By comparing color words and fruit words under match and mismatch conditions, the experiment examined how word meaning and object-related color knowledge influence reaction time and accuracy during font color judgment. Participants generally performed better in match conditions than in mismatch conditions, suggesting that both color-word meaning and object-word meaning can produce Stroop-like interference when they conflict with the displayed font color. Notably, the Fruit-Mismatch condition showed a larger decrease in accuracy but shorter reaction times than the Color-Mismatch condition, suggesting that participants may have been rapidly influenced by typical fruit-color associations and responded more quickly but less accurately. Overall, these findings support the idea that familiar object-color knowledge can be automatically activated and interfere with visual color judgment in a way similar to the traditional Stroop effect.

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